

The role of microtubules in processive kinesin movement

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Kinesins are microtubule-based motors that are important for various intracellular transport processes. To understand the mechanism of kinesin movement, X-ray crystallography has been used to study the atomic structures of kinesin. However, as crystal structures of kinesin alone accumulate, it is becoming clear that kinesin structures should also be investigated with the microtubule to understand the contribution of the microtubule track to the nucleotide-induced conformational changes of kinesin. Recently, several high-resolution structures of kinesin with microtubules were obtained using cryo-electron microscopy. Comparison with X-ray crystallographic structures revealed the importance of the microtubule in determining the conformation of kinesin. Together with recent biophysical data, we describe different structural models of processive kinesin movement and provide a framework for future experiments.

Introduction

Kinesin is a microtubule-based motor that uses ATP as the energy source. Kinesins are involved in various intracellular trafficking events such as axonal transport and chromosome segregation [1,2]. Several models have been proposed to explain the movement of conventional kinesin, such as symmetric hand-over-hand, asymmetric hand-over-hand and inchworm. Recent experiments have focused on distinguishing these models, and recent biophysical experiments favor the asymmetric hand-over hand model [3,4]. However, the track on which kinesin moves, the microtubule, is often considered as a passive track in these models. However, recent structural data derived from cryo-electron microscopy (cryo-EM) indicate that microtubules have a more active role. Here, we describe recent structural data and the current problems surrounding understanding the mechanism of kinesin movement, and discuss the role of kinesin–microtubule interactions.

The kinesin motor domain is similar to switching G proteins

The motor domain of kinesin is of great interest because it can efficiently convert chemical energy from ATP hydrolysis to mechanical movement (for the overall architecture of kinesins, see Box 1). Kinesin can generate 8-nm steps against a load of 7 pN using ~100 pN nm of chemical energy derived from hydrolysis of one ATP molecule (Here, we use pico-Newton nanometer, pN nm, per molecule as

units of free energy according to Howard [5].) Therefore, ~50 pN nm of mechanical energy is generated and the energy conversion efficiency is close to 50%. To understand the highly efficient conversion mechanism, the structure and functions of the kinesin motor domains have been extensively studied. More than a decade ago, X-ray crystallography revealed that the structure of the kinesin motor domain is similar to that of myosins and GTP-binding proteins (G proteins) [6,7]. In G proteins, the ‘switch region’ can detect the presence (GTP) or absence (GDP) of γ -phosphate in the nucleotide and respond by undergoing conformational changes.

Is nucleotide state enough to ‘switch’ kinesin conformation?

As expected from the similarity between kinesins and G proteins, marked conformational changes have been observed within the ‘switch regions’ of kinesin motors using X-ray crystallography [8,9]. The first switch region consists of a single loop, L9, called the switch I loop [Figure 1a(ii)]. Like G proteins, this loop is close to the phosphate of the bound nucleotide and presumed to sense the presence (ATP) or absence (ADP) of a γ -phosphate. A second conformational change is observed around the switch II cluster that surrounds helix $\alpha 4$ [Figure 1a(ii)]. In plus-end directed motors, the movement of the kinesin switch II cluster is always correlated with the conformation of the third region, the neck linker [Figure 1a(ii)]. The neck linker is docked onto the kinesin catalytic core when the switch II helix is closer to the nucleotide [Figure 1a–c(ii)], whereas it is undocked or invisible when the switch II helix is closer to the β -sheet and in the same plane as the $\alpha 6$ helix [Figure 1a,b(i)]. The docking of the neck linker on the kinesin catalytic core presumably delivers a plus-end-directed pull on its cargo [8,10,11] (Figure 2). This model is called ‘the neck-linker-docking model’.

A straightforward interpretation of the X-ray crystallographic structures of kinesin is that the energy-transmitting pathway starts from the γ -phosphate of the nucleotide, goes through the switch I loop and switch II helix, and reaches the neck linker. However, accumulating crystallographic data indicate that the nucleotide alone might not be enough to induce conformational switching. First, the conformations of switch I do not depend on nucleotide state. Second, a so-called ‘ATP-like’ structure has been observed in the presence of ADP [Figure 1c(ii)]. Similarly, an ‘ADP-like’ structure has also been observed in the presence of an ATP analog [12] [Figure 1c(i)]. Furthermore,

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Box 1. Architecture of kinesins

Conventional kinesin consists of two heavy chains and two light chains that form a dumbbell-shaped structure (Figure 1a). One kinesin molecule has two heads, a long coiled-coil stalk and a cargo-binding tail with light chains [53,54] (Figure 1b). The head domain, known as the 'motor' domain, has both ATPase- and microtubule-binding activities. The two heads are connected to the α -helical coiled-coil that forms the stalk domain (Figure 1c). The beginning of the stalk is called the 'neck' and a short (13 amino acid) polypeptide segment, the 'neck linker', connects the head and the neck (Figure 1d).

According to the standardized nomenclature, kinesins are classified into 14 families [55] (Figure 1e), and domain organizations have been deduced using the SMART server [56]. Plus-end-directed kinesins have an N-terminal motor domain(s) and include a dimeric kinesin heavy chain (KHC; standardized name, Kinesin-1), monomeric KIF1A (Kinesin-3), Eg5 (Kinesin-5) and CENP-E (Kinesin-7). KIF2C (Kinesin-13) is a microtubule-depolymerizing kinesin, which has a centrally located kinesin motor domain. Minus-end-directed kinesins belong to the Kinesin-14 family, and include Ncd, Kar3 and kinesin-like calmodulin-binding protein (KCBP), and their motor domains are located at the C-terminal region.

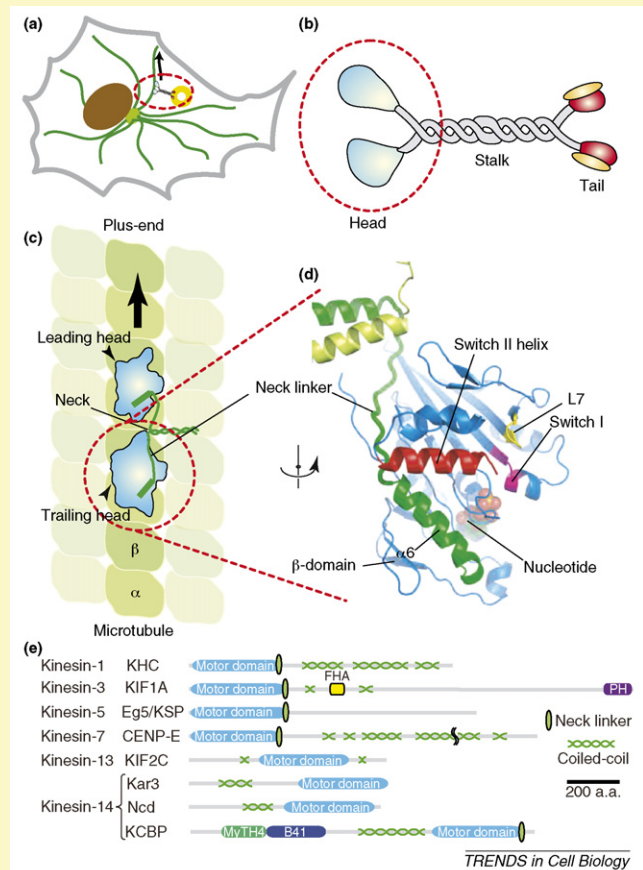


Figure 1. Architecture of kinesins. (a) Kinesins are involved in various cargo transports in cells. (b) Schematics of the overall structure and (c) two-head-bound state of conventional kinesin are shown. (d) The atomic structure of kinesin head viewed from the microtubule binding surface. Domains involved in the conformational changes are labeled. (e) The primary domain organization of kinesins for which crystal structures are available.

recent extensive multivariate analyses of 37 available atomic structures from five different kinesin subfamilies identified the switch II cluster (α 4-loop12- α 5-loop13) as the first principle component of conformational differences, but failed to support tight coupling between nucleotide state and kinesin conformation [13]. In summary, there are

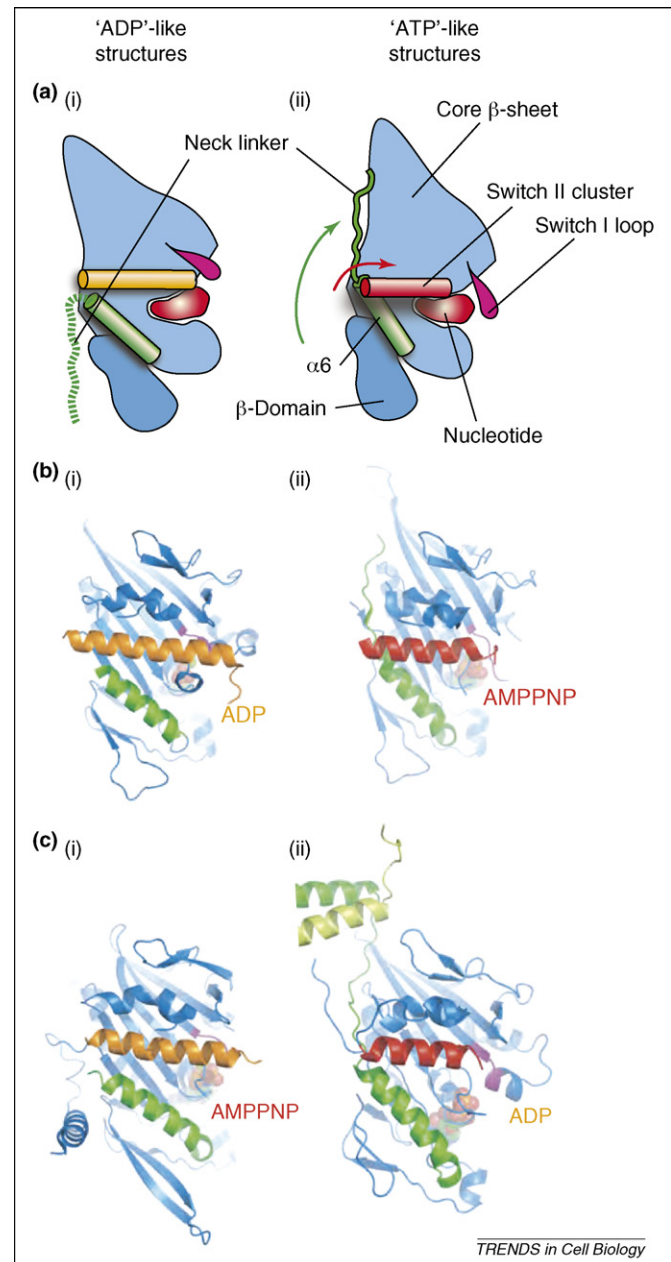


Figure 1. Is nucleotide state enough to switch the conformation of kinesin? Kinesin structures are approximately classified into two groups: so called 'ADP'-like and 'ATP'-like structures. Contrary to the names, these two structures are not always correlated with the bound nucleotide in the X-ray crystallographic structures. (a) Schematic structures of the kinesin motor domain. The ADP-like (i) and ATP-like (ii) structures are distinguished by the relative arrangement between the switch II cluster and the core β -sheet. In the ADP-like structure, the switch II cluster is closer to the β -sheet and in the same plane as the α 6 helix. In the ATP-like structure, the helix moves towards the nucleotide like a 'piston' and the distance from the core β -sheet is relatively greater. In most 'ADP'-like crystal structures, the switch II helix is longer than that of the 'ATP'-like structures; however, this might not be true in the microtubule-bound state because a longer switch II helix has also been observed in the AMPPNP state [25]. (b) Examples of crystal structures of kinesin. (i) KIF1A (monomeric plus-end-directed kinesin) in the ADP state (PDB: 1I5S [8]). (ii) KIF1A in the AMPPNP state (PDB: 1VFV [16]). (c) Examples of kinesins, the structures of which do not match with the bound nucleotide: (i) KIF2C (PDB: 1V8K [12]) with bound AMPPNP, crystallized in the ADP-like form. (ii) Rat conventional kinesin with ADP (PDB: 2KIN [14]), crystallized in the ATP-like form.

two principle conformations of the switch II cluster in solution and crystals, but they are interchangeable in solution and independent of nucleotide state [9].

Examination of the available atomic structures of kinesins implies that the pathway that induces conformational

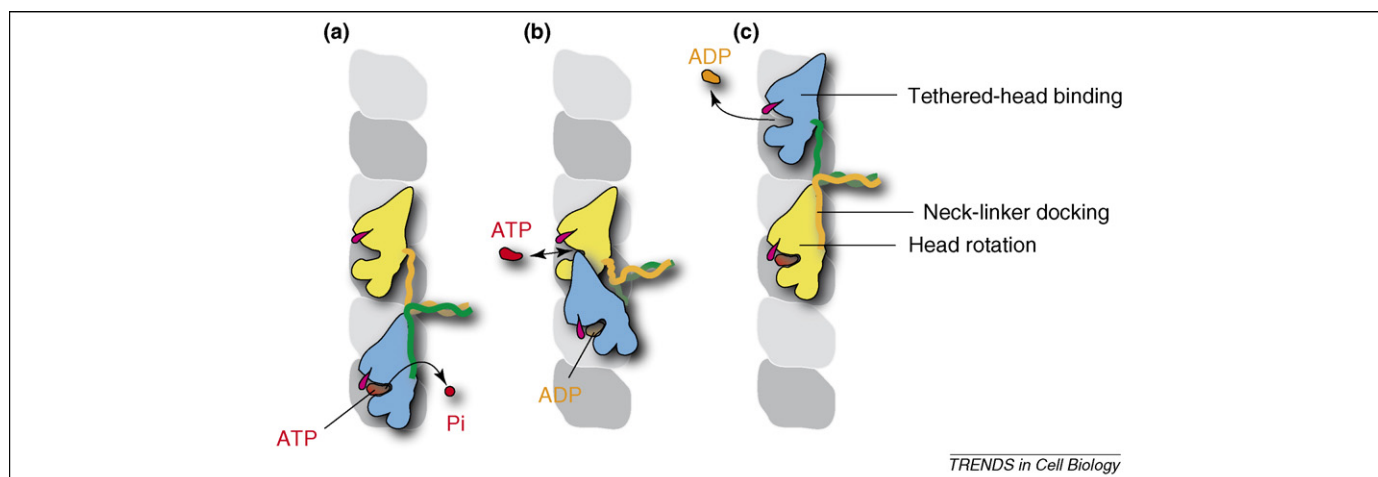


Figure 2. A model for kinesin stepping. (a) The two kinesin heads are bound to a tubulin heterodimer along a microtubule protofilament. The trailing head (blue) hydrolyzes the ATP and dissociates from the microtubule. The head releases phosphate and is now in the ADP-state. (b) The head is tethered to the microtubule-bound head (yellow) by the neck linker. Under a small load, the tethered head is 'parked' behind the bound head. The tethered head could go back to the previous backward binding site under a higher load, but the head cannot accommodate the tilted conformation because of the limited length of the neck linker. Because the tilted conformation is required for the release of ADP, the tethered head remains in a weak binding state. While the trailing head is in the weak binding state, the bound head (yellow) sometimes binds to ATP but, because the nucleotide-binding pocket is open, the head cannot hydrolyze ATP, thus premature dissociation of kinesin from microtubules is prevented. (c) ATP binding to the bound head induces docking of the neck linker and rotation of the head. These two conformational changes also enable the tethered head to bind to the forward binding site, where it can take the tilted conformation, release ADP and switch from a weak to a strong microtubule-binding state.

change might work backwards, with the docking state of the neck linker controlling the conformation of the switch II helix. The ATP-like structure is observed only when the C-terminal neck linker is in the docked state. No ATP-like structure has been crystallized for kinesins that lack the C-terminal neck linker (e.g. Kar3, Ncd, Vik1 and KIF2). By contrast, all kinesin subfamilies with neck linkers (e.g. conventional kinesins [9,14,15], KIF1A [8,16], Eg5/KSP [17], CENP-E [18] and KCBP [19]) have been crystallized in the ATP form, when the neck linker is docked to the core. Therefore, the docking and undocking of the neck linker might significantly influence the ATPase cycle of kinesin. The results of recent biochemical experiments also support this idea by showing that the ATPase activity of the kinesin motor domain is significantly lower if the neck linker is locked to the core by cross-linking [20].

The microtubule 'track' is essential for conformational switching

As discussed, X-ray crystallographic analyses have provided two principle conformations for the kinesin motor domain but have failed to show the tight coupling between nucleotide state and conformation. To investigate kinesin conformation in the presence of microtubules, several groups have analyzed high-resolution structures of kinesin-microtubule complexes using cryo-EM at a resolution <20 Å [21,22] with some at ~10 Å with α helices visualized in the density maps [23–25]. In addition, these cryo-EM studies include two kinesin structures in the absence of nucleotide (i.e. for Kar3 and human kinesin), which have not been obtained by X-ray crystallography and represent the missing piece in the ATPase cycle of kinesin.

The structures of the kinesin-microtubule complex are approximately divided into three groups based on the angle of the core β -sheet relative to the microtubule. The core β -sheet of Kar3, in all nucleotide states, is the most tilted [Figure 3a,b(i)]. The core β -sheets KIF1A in the ADP state and human kinesin in the no-nucleotide state

are tilted so that the plus-end tip of the β -sheet points towards the 1'o clock position [Figure 3a,b(ii)]. The core β -sheets of KIF1A and CENP-E in the AMPPNP state are in the 'upright' conformation, whereby the plus-end tip of the kinesin core points towards the 12'o clock position [Figure 3a,b(iii)]. In summary, the orientation of the core seems to be tightly coupled with the bound nucleotide in the presence of microtubules, except for the minus-end directed kinesin Kar3.

This conformational change was first observed as ~20° rotation in low-resolution cryo-EM structures of a KIF1A-microtubule complex [25], but it has been controversial [26]. The rotation seems smaller (5° and 12°, respectively) in the UNC104 (the *Caenorhabditis elegans* homolog of KIF1A)-microtubule complex [27] and in the rat conventional kinesin-microtubule complex [28]. According to the new high-resolution studies on three plus-end-directed kinesins (Figure 3), the rotation angle seen from the outside of the microtubule is ~11°. These latter three values are approximately consistent, and it is reasonable to conclude that rotation is a generic conformational change among plus-end-directed kinesins. However, because not all nucleotide states of the kinesin-microtubule complex are available at high-resolution for one species of plus-end-directed kinesin, future studies are required.

What is the origin of the rotation? There are two X-ray crystallographic conformations for the switch II helix. In the kinesin-microtubule complexes, the switch II helix is at the center of the microtubule-binding surface. Therefore, the core rotates while the switch II helix is fixed to the microtubule. In the 10-Å density maps, the switch II helices are visible and support this idea [23–25]. In addition, the length of the switch II helices according to the density maps are different from that determined by the crystal structures, indicating that the flanking loops are involved in microtubule sensing together with switch I loop.

It is also interesting that new open and closed conformations of the nucleotide-binding pocket are observed in

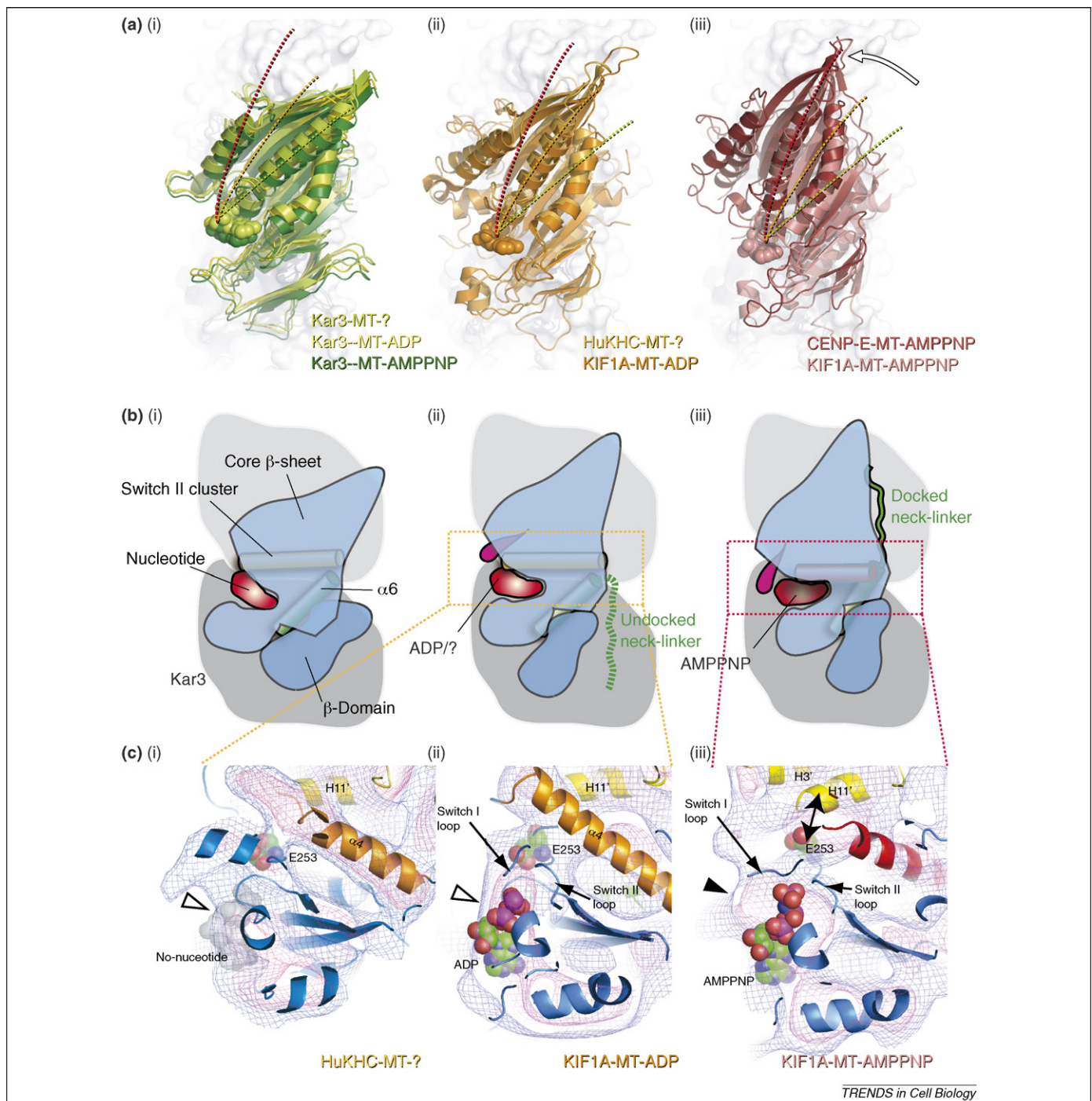


Figure 3. Atomic models of kinesin-microtubule complexes derived by fitting crystal structures into cryo-EM density maps. **(a)** Atomic models of the kinesin-microtubule complex. (i) Kar3-microtubule complexes in the no nucleotide (green), ADP-bound (yellow) and AMPPNP-bound states (dark green). (ii) The human conventional kinesin-microtubule complex in the no-nucleotide state (light orange; PDB code: 2P4N) [24]. The KIF1A-microtubule complex in the ADP-bound state (dark orange; PDB code: 2HXH) [25]. (iii) The CENP-E-microtubule complex in the AMPPNP-bound state (red) [21]. The KIF1A-microtubule complex in the AMPPNP-bound state (pink; PDB code: 2HXF) [25]. To compare the orientations of the kinesin head, the $\alpha 2$ helix is traced by broken lines. The microtubules are shown as white surface with their plus-end up. Note that these atomic models show only the overall structures of the complexes; structural changes found by cryo-EM studies are not incorporated. **(b)** Schematic models of the kinesin-microtubule complex, each corresponding to the atomic model above. The schematic models are flipped compared with Figure 1(a). **(c)** Electron-density maps of the kinesin-microtubule complex surrounding the nucleotide-binding pocket, showing (i) human conventional kinesin in the no-nucleotide state, (ii) KIF1A in the ADP-bound state and (iii) KIF1A in the AMPPNP-bound state.

the cryo-EM density maps. The nucleotide-binding pocket is open in the ADP-state KIF1A [25] [Figure 3c(ii)] and in the no-nucleotide state of human kinesin [24] [Figure 3c(i)]. In the Kar3-microtubule complexes, the pocket is wide open in the no-nucleotide state, and partially open in the ADP state [23]. In both Kar3 and KIF1A, the nucleotide-binding pocket is closed and forms a 'phosphate tube'

around the triphosphate in the ATP state [23,25] [Figure 3c(iii)]. The opening and closing of the nucleotide-binding pocket is probably caused by the conformational changes of the switch I loop because electron paramagnetic resonance (EPR) probes covalently bound to the loop have shown changes in mobility between ADP- and ATP-analog states [29]. In addition, it has also been

demonstrated that the mobility of EPR-labeled ADP analogs decreases when kinesin binds to microtubules [30]. The latter result is somewhat perplexing because the cryo-EM results indicate that the nucleotide-binding pocket is open, whereas the EPR study indicates that the pocket is closed.

Unexpectedly, the structural changes mentioned are not always the same between plus-end-directed kinesins and the minus-end-directed motor Kar3 [23]. First, the rotation of the Kar3 core does not differ greatly between the different nucleotide states, but is more tilted compared with other kinesins [Figure 3a,b(i)]. This might be owing to the fact that Kar3 does not have a neck linker. Because a 70° rotation of the N-terminal coiled-coil was observed using cryo-EM for Ncd [31], the N-terminal neck might be necessary to stabilize large structural changes.

Second, in the no-nucleotide state, the switch II helix of Kar3 seems 'melted' and is no longer visible in the density map as a long helical structure, whereas the helix seems extended and clearly visible in human kinesin [Figure 3c(i)]. Third, L7, which faces the microtubule and is close to switch I [Figure 1c(ii)], together with flanking β -sheets, moves towards the microtubule in the no-nucleotide state. These differences might be owing to the difference between plus-end-directed and minus-end-directed motors. It should be noted that the resolution of the Kar3 density map in the no-nucleotide state is lower than that of human kinesin. If a Fourier shell correlation cut-off of 0.5 – a measure of reproducibility of the cryo-EM density map – is used to define the resolutions are 9.1 Å for human kinesin and 18 Å for Kar3 (see Ref. [23]).

From monomer to processive dimer

A unique characteristic of conventional kinesin is that it can move more than a few microns along a microtubule without detaching, which is called 'mechanical processivity'. It is widely accepted that this processivity is enabled via communication between the two motor domains of kinesin [4,32,33]. For example, the detachment of the kinesin head is accelerated when a forward strain is applied to the head. This is one of several strain-dependent control points that are referred to as a 'gates' [33]. During kinesin movement, the gate 'opens' for the trailing head only when both kinesin heads are bound to the microtubule (two-head-bound state); this is required for the forward strain, ensuring at least one head is bound to the microtubule at any given time.

To understand the mechanism of gating, structural information about the two-head-bound state of kinesin is essential. An EM study using the unidirectional metal shadowing technique on SH3-labeled dimeric kinesin demonstrated that the two heads bind to two consecutive tubulin dimers along one protofilament [34]. This result is consistent with a study using fluorescence polarization microscopy [35]. However, the detailed structure is yet to be determined. In the following, we try to extrapolate a model for the structure of the two-head-bound state from the data available.

First, we hypothesize that strain is applied to the neck linker in the two-head-bound state, judging from the

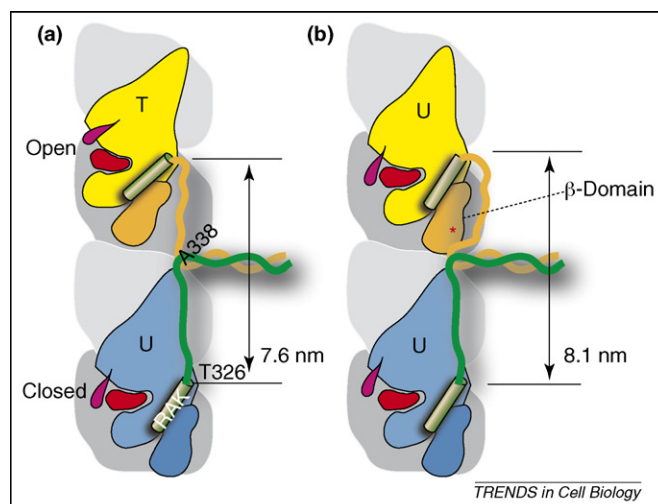


Figure 4. A structural model of kinesin processivity. To extrapolate a model for the two-head bound state, the upright (U) and tilted (T) conformations of the monomeric kinesin-microtubule complex structures obtained from cryo-EM are placed along a microtubule protofilaments. There are four possible dimer conformations (X-Y, whereby X is the leading head and Y is the trailing head): T-U, T-T, U-U and U-T. (a) Among these possible conformations, the distance between the start of the neck linker is the shortest in the T-U configuration because its Thr326 is 0.5 nm closer to the minus-end of the microtubule in the tilted conformation. The two ends of the neck linker are also shown based on the rat conventional kinesin sequence. (b) Other variations result in longer distances (>81 Å). If the leading head is in an upright conformation, the β -domain clashes with the extended neck linker.

length of the neck linker. The neck linker is known to start at Thr326 in rat conventional kinesin (Figure 4a) because a structural comparison between the kinesin (PDB code: 3KIN [14]) and Eg5 (PDB code: 1II6 [36]) reveals that, from Thr326, the conformation deviates. The C-terminal end of helix α 6, including a well-conserved Arg323-Ala-Lys325 sequence, does not change its conformation between the two crystal structures. Probably, the long side-chain of lysine prevents excessive unzipping of the neck linker. The neck linker ends at the junction with the α -helical coiled-coil (Ala338; Figure 4a). Chemical cross-linking of two kinesin molecules at the equivalent position does not affect the processivity of kinesin, indicating that unwinding of the α -helical coiled-coil is not necessary [37]. Taking together, these data show that the neck linker is ~13 amino acids for each head in the conventional kinesin.

In the two-head-bound state, a pair of 13-amino-acid neck linkers extends to ~81 Å to span two tubulin dimers (Figure 4). Within each neck linker, each amino acid is extended to, on average, 3.1 Å, which is close to the fully extended state of polypeptide. Therefore, the neck linker must have a certain amount of strain in it. A probable result of this strain is that the leading head is tilted, similar to the ADP/no-nucleotide state, and the trailing head is upright, similar to the AMPPNP state (Figure 4a). This model was originally proposed by Sablin and Fletterick [32] but we have refined it by incorporating the new cryo-EM data. This configuration is preferred for the following reasons. First, when the leading head is in the upright conformation, a small β -domain lobe clashes with the neck linker [32] (Figure 4b). Second, it is also reasonable that the head is rotated clockwise by the strain applied

to the right side of the kinesin head. If ~ 17 pN nm of strain is applied to the neck linker [38], a large amount of free energy is required for only a small extension caused by core rotation. For example, an extra 6–8 pN nm (1.5–2 kT) of energy is required to extend the neck linker from 7.6 nm (Figure 4a) to 8.1 nm (Figure 4b). Collectively, considering the large unfavorable entropy changes associated with extension of the neck linker, the configuration in which the leading head is tilted and the trailing head is upright is a favorable configuration (Figure 4a).

This configuration, in combination with the high-resolution structures of kinesin–microtubule complexes, leads to a structural model of processivity. In both the KIF1A–microtubule complex in the ADP state and human kinesin–microtubule complex in no-nucleotide state, the nucleotide-binding pocket of kinesin is open, which is not observed in the crystal structure. This indicates that microtubules work as nucleotide-exchange factors for the tilted form of kinesin. Although tilting of the leading head has not been shown experimentally, the lower affinity to nucleotide under a minus-end-directed load has been demonstrated by measuring the unbinding force at various ADP concentrations [39]. As a result of the open nucleotide-binding pocket at the leading head, simultaneous detachment of the two heads from microtubules is prevented, thereby ensuring kinesin processivity. This theory is also supported by results of a biochemical experiment that show that elongation of the neck linker reduces kinetic processivity of kinesin [40]. In the kinesin mutants, it is likely that both heads can take the upright conformation and simultaneously hydrolyze ATP.

Which mechanism mainly generates the force?

The neck-linker-docking model is very attractive. In fact, recent single-molecule fluorescence resonance energy transfer (FRET) analysis has clearly shown that ATP binding triggers the transition from ‘waiting state’ to the two-head-bound state, presumably, by inducing the docking of the neck linker [41,42]. In the waiting state, the tethered head is close to the microtubule-bound head (Figure 2b). The result is also consistent with a recent biochemical study that demonstrated that only one unpolymerized tubulin heterodimer binds to one dimeric kinesin in the absence of nucleotide, whereas one dimeric kinesin binds to two tubulin heterodimers in the presence of an ATP analog [43]. Probably, in this waiting state, the microtubule-binding surface of the tethered head is shielded by the microtubule-bound head. Taken together, neck-linker docking is one of the major conformational changes that occur during the stepping of kinesin.

As discussed elsewhere (see Refs [3,4,44]), one persistent problem is that the free energy change associated with neck-linker docking is ~ 5 pN nm [45], which is only $\sim 10\%$ of the energy required to generate an 8-nm step against a 7-pN load. If neck-linker docking is the only force-generating mechanism, under a 1-pN load, the neck linker would be in equilibrium between docked and undocked states, and kinesin would move in forward and backward directions equally. This indicates that other force-generating mechanisms must be involved.

One potential mechanism that contributes to force generation is the binding of the tethered head to the forward-binding site. This has also been proposed previously [45], although the previous proposal did not clearly mention why the tethered head preferentially binds to the forward-binding site even under a >1 -pN load. Is it possible for the tethered head to distinguish between forward- and backward-binding sites? The answer seems to be ‘yes’ because single-headed kinesin, the neck linker of which is tethered to the optically trapped bead, preferentially binds to the forward site [46,47].

What is the structural basis of this forward bias? In both single-headed and two-headed kinesins, the kinesin heads are connected by the neck linkers and their binding to the forward site induces a tilted conformation. As discussed, if the tilted conformation is required to open the nucleotide-binding pocket, ADP release and subsequent strong binding to the microtubule only happens at the forward-binding site. In this case, ADP release is the force-generation step. Because the free-energy change associated with binding of a kinesin-motor domain to microtubules is large (>50 pN nm [48]), this mechanism would provide enough free-energy to support kinesin stepping even under a high load.

It has been demonstrated experimentally that the binding of the tethered head in the forward position is entropically favored over binding to the trailing position by 16 pN nm based on measurements of temperature-dependence of forward and backward stepping rates [49]. It has also been proposed that ADP release is a force-producing event of the kinesin cycle based on the analysis of a kinesin mutant that shows a high frequency of rapid double 8-nm steps at high load [50].

The interaction between the kinesin head and microtubule might also drive kinesin forward by the asymmetric structural changes of the microtubule protofilament [51]. A recent 17-Å resolution cryo-EM study identified a significant structural difference in the α -tubulin compared with plain and kinesin-decorated microtubules [22]. The structural changes include continuous axial inter-dimer contact in the kinesin-decorated microtubules, indicating that structural changes might propagate along the protofilament, generating asymmetric potential. Although no similar structural changes have been observed in other cryo-EM studies [23–25], higher-resolution studies might be required to detect subtle structural changes induced by kinesin decoration.

The other possible force-generating mechanism is the rotation of the kinesin head (Figure 3). This head rotation brings the root of the neck linker 0.5 nm closer to the microtubule plus-end. Although the displacement is small compared with the 8-nm step size of kinesin, the rotation can potentially generate force if the torque is large. However, the small rotation is observed only by cryo-EM and needs to be investigated by other methods.

Concluding remarks and future perspectives

Here, the structural aspects of kinesin are the focus of review. With the new structures of kinesin–microtubule complexes, it has become clear that microtubules are important for tight coupling between the kinesin and

nucleotide. However, we think there are two fundamental problems which remain unsolved:

First, the interaction between a kinesin and microtubule is not understood at the atomic level because of the difficulty in crystallizing kinesin–microtubule complexes. Consequently, the structural basis of how the microtubule activates ADP release or ATP hydrolysis by kinesin is not well understood. Recent advances in mutating tubulin [52] in addition to the availability higher-resolution structures of kinesin–microtubule complexes using cryo-EM will eventually give more concrete answers to the question.

Second, because the cryo-EM and X-ray crystallographic structures are static, it is not clear whether the observed conformational changes by these methods represent key structures during the force generation of kinesin. In addition, the contribution of each conformational change to force generation is not clear. Future studies should determine how much each mechanism (e.g. neck-linker docking versus tethered-head binding) contributes to the force generation in kinesin. For such quantitative studies, new techniques such as single-molecule FRET [41] and/or fluorescence polarization microscopy [35] can be used to observe movement and conformational changes simultaneously.

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